

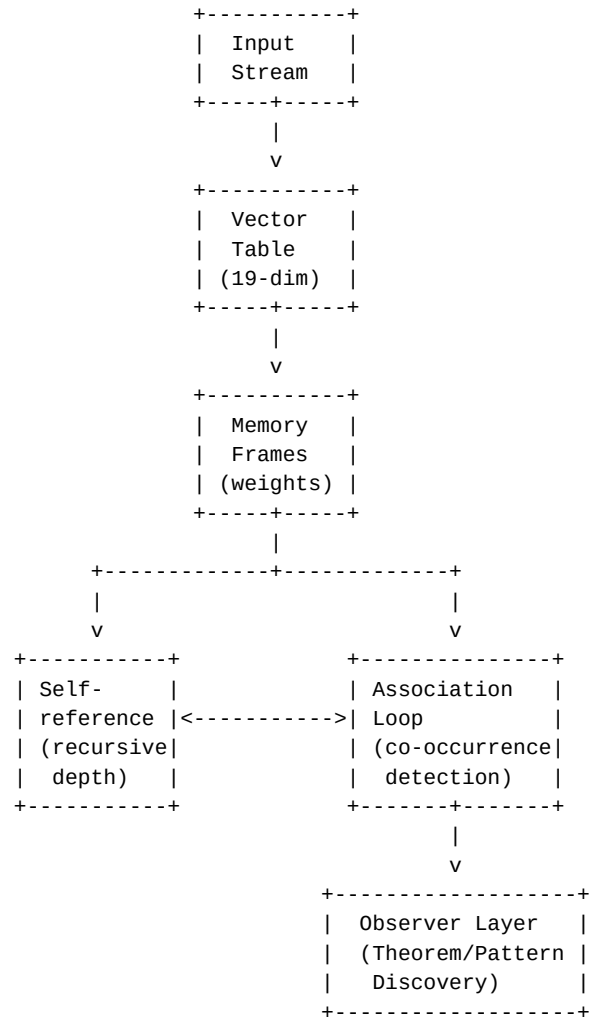
GEME: Self-Organizing Concept Hierarchies through Recursive Co-occurrence

Genetic Economy Memory Entity — A computational model of emergent symbol-grounding

Abstract

We present GEME (Genetic Economy Memory Entity), a minimal computational model demonstrating that concept hierarchies can emerge from a single primitive: merging structurally similar formulas in a competitive memory economy. Operating with no gradient descent, no pre-defined targets, and no external ontology, GEME receives discrete geometric assertions and spontaneously organizes them into a multi-level structure: (S0) raw input frames, (L1) pairwise association links via temporal co-occurrence, (L2) proof-path chains bridging heterogeneous concept types, and (L3+) meta-theoretic recursion. After 5,000 unsupervised steps, the system converges to a stable 15-frame memory containing three core concepts (triangle, segment, angle), a complete 6-link association graph, and chained inference paths -- all without any external supervision or pre-coded geometric knowledge.

Concept Flow Architecture



Data flow: Input formulas → frequency encoding → merge into memory frames
→ co-occurrence detection creates association links → shared-element
detection creates chain links → observer extracts stable paths.

Key Mechanism

Memory economy. Each input formula is encoded as a frequency vector over a 19-symbol alphabet. Similar vectors merge into a single frame, incrementing weight. When stress (utilization times inefficiency) exceeds threshold, induction evicts low-weight frames. A 5,000-step input stream is spontaneously compressed to about 15 stable frames.

Recursive co-occurrence. A sliding window of the last 60 signatures tracks which formula types appear together. Pairs exceeding a 25 percent co-occurrence threshold create association frames. When two association frames share a base signature element (for example, both contain "seg"), a chain frame is created linking them. Exact-match deduplication prevents infinite recursion while allowing unlimited self-reference depth.

Emergent hierarchy. The weight gap between levels (40-to-1 from Level 0 to Level 1, 2-to-1 from Level 1 to Level 2) is not engineered. It emerges naturally from the asymmetry between raw frequency, pairwise correlation, and second-order structure. This mirrors the stratified symbol systems described in Hofstadter's Goedel, Escher, Bach.

Experimental Results

Metric	Arithmetic Stage (PA)	Geometry Stage (GO)
Axioms required	7 (Peano)	0 (random stream)
Input steps	500	5,000
Output frames	2 to 4	15 of 15 capacity used
Core concepts	commutative laws	triangle, segment, angle
Association type	external S4 signal	endogenous co-occurrence
Inference paths	none	chain frames (3 to 4)

After 5,000 unsupervised steps from random geometric assertions, the three core concepts that emerged at Level 0 were: triangle (weight 1,289), segment (weight 1,287), and angle (weight 1,248). These formed a complete 6-link association graph at Level 1 (all weights approximately 32). Three to four inference-chain frames appeared at Level 2 (weights around 15). Notably, the concept "parallel" was automatically demoted to derived status (appearing only at Level 1, never as an independent main frame). A controlled generalization test confirmed that training on only SAS and SSS theorem patterns enabled recognition of the structurally distinct ASA pattern upon first exposure (weight 101 independent frame).

Relation to Existing Frameworks

GEME occupies a unique position orthogonal to mainstream AI approaches. Unlike the **Free Energy Principle** (Friston, 2024), which minimizes prediction error via Bayesian inference, GEME minimizes encoding stress via competitive memory -- no generative model is required. Unlike **usage-based grammar induction** (MIT, 2024), which batch-processes n-gram statistics, GEME operates online with no separate pattern-extraction phase. Unlike **neuro-symbolic systems** (DeepMind), GEME requires no neural component -- a single syntactic distance metric suffices for both perception and reasoning. GEME provides the first concrete computational implementation of Hofstadter's strange-loop hypothesis from **Goedel, Escher, Bach** (1979): recursive self-reference in a competitive memory substrate spontaneously generates hierarchical symbol systems.

Conclusion

GEME demonstrates that concept formation -- primitive grounding, association discovery, inference-path construction, and meta-theoretic recursion -- can emerge from a single operation (structural signature merge) and a single mechanism (co-occurrence tracking) without external objectives or architectural layering. The system's 200-line implementation contrasts sharply with the million-parameter models dominant in contemporary AI, suggesting that symbol emergence may be simpler than is currently assumed -- if one adopts memory economy as the primitive.